

Ultracapacitors: Another no-maintenance alternative?

Two-wheelers are fitted with lead-acid battery to meet intermittent loads such as horn, direction indicators and brake light. These loads are also related to safety of driver. Typically, life of the battery is about two years if maintained properly. In practice, most users do not replace the factory-fitted battery once its life is over and continue to use the vehicle compromising safety features.

Ultracapacitors, with their short charging time, are ideally suited for these intermittent loads and have a usable life exceeding that of vehicle. Experiments conducted on entry-level motorcycles clearly indicate that the ultracapacitor module adequately meets the electrical requirements of the vehicle and can be directly substituted for the battery.

Advantages are long life, weight and space reduction, no-maintenance and lead-free environment-friendly product.

Ultracapacitors store energy electrostatically. These can be charged or discharged very fast, in seconds. Table I shows the advantages and disadvantages of ultracapacitors and batteries.

Following are the potential areas of application where ultracapacitors' strength can be utilised:

1. Automotive applications and electric vehicles
 - Regenerative braking to improve fuel efficiency
 - Electric vehicles to enhance performance and efficiency
 - Delivery vans with > 100 start/stops and short travel distance between stops
2. All-weather quick-start applications
 - Instant starting in cars even in severe winter conditions
 - Downsizing and life extension of batteries
3. No maintenance applications
 - Buoys in sea-lanes
 - Construction sites and road hazards
 - Genset starters, emergency/safety lighting, windmills, solar systems and medical equipment
4. Quick-charge applications
 - Cordless power tools such as hand-drill machines and tap-runners
5. Memory back-up applications
 - Consumer electronics to protect/lock data for life

—Vijay Chhedha and D.Y. Vernekar, Chhedha Electricals & Electronics, Pune

TABLE I
Advantages and Disadvantages of Ultracapacitors and Batteries

Ultracapacitor	Battery
Advantages High expected lifetime of more than 10 years, charge-discharge life cycle of >500,000 cycles, high power density, short charge-discharge time, maintenance-free (fit-and-forget device), wide operating temperature range (-40°C to +70°C), completely dischargeable, less weight, environment friendly, damage-proof against short circuit, no special charge-control needed	High energy density, low self-discharge, flat discharge curve
Disadvantages Low energy density, high self-discharge	Long charging times, lifetime of about 1 to 3 years, charge discharge life of about 1000 cycles, regular maintenance needed, charging-control necessary, more weight, over-scaling needed for some applications

TABLE II
Comparison of Capacitors, Ultracapacitors and Batteries

Parameter	Unit	Capacitor	Ultracapacitor	Battery
Specific energy	Wh/kg	0.1	3 to 5	80 to 100
Specific power	W/kg	10 ⁶	4000	300
Time to charge	Sec	10 ⁻² to 10 ⁻⁴	0.3 to 15	>1000
Time to discharge	Sec	10 ⁻² to 10 ⁻⁴	0.3 to 15	1000 to 10,000
Cycle life	Cycle	100 × 10 ⁶	0.5 × 10 ⁶	1000
Life time	Year	25	20	1 to 3
Efficiency	Per cent	>95	85-98	70-85

2V 3000 Ah

"Previously, only one type was available in the Indian market, but now

we can offer these batteries ideally designed for different applications, such as gel-based for preventing water loss

and deep-cycle for solar applications," says Praveen Kumar Sood, chairman, Regnant.

Manufacturing in India

India is an important market with a GDP growth of 7 to 8 per cent year-on-year. With rapid growth in the power and energy sector, all existing battery players are emphasising on expansion.

"With the rising production costs overseas, India is definitely a good destination if the right skills and the plate chemistry technologies can be mastered indigenously. The large, technically qualified, scientific workforce available in India can be used, provided necessary productivity of our employees can be developed to the level of other major players worldwide. As far as raw material is concerned, India will continue to be a net importer of lead (a major raw material), as our domestic production is too less and also quality is not up to the world standards. For SMF VRLA, the lead purity required is of the order of 99.9999 per cent. A major cost in battery production is the electricity cost in charging-discharging and plate-formation processes. Therefore abundant and economical supply of electric power can be a constraint," informs Sood of Regnant.

"The challenge for manufacturing in India largely centres around the availability of lead. While lead is mined in India, India's share was only 1 per cent in lead with metal production of 454,000 MT in 2000. By 2016, the demand-supply gap of lead is estimated at 164,200 MT. This has led to a flourishing unorganised market for recycled lead, which is used by the unorganised sector to manufacture and sell batteries at low costs," says Biju Bruno, managing director, GreenVision Technologies.

A few years ago, The Tamil Nadu Pollution Control Board had threatened withdrawal of the pollution control certificate to a battery-manufacturing unit in the State for the company's alleged non-compliance to battery